INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO

Introduction to Cryptography

Lab 03: Permutation cipher *March 5, 2026*

Please solve the following exercises using C/C++, C#, Java or Python to develop your source code, but use only one of them to do all the exercises.

1. Programming exercises I

Do the following exercises by your own. **You must upload photo or image of your pseudocode, screenshots of your source code and screenshots of your programs running to Microsoft Teams today, at the end of the session**

1. Write by hand in your notebook or tablet, the pseudocode of an algorithm to randomly generate permutation π. You must receive as input the size of the permutation $n \geq 3$, the output must be the permutation π of size n. Follow the style given in the file pseudocode_examples.pdf
2. Design a function to implement the previous algorithm. Your function must receive as a parameter an integer $n \geq 3$ and return the permutation π.
3. Write by hand in your notebook or tablet, the pseudocode of an algorithm that receive as input a permutation π of size $n \geq 3$, the output must be the inverse permutation π^{-1}. Follow the style given in the file pseudocode_examples.pdf
4. Design a function to implement the previous algorithm. Your function must receive as a parameter a permutation π of size $n \geq 3$ and return the inverse permutation π^{-1}.

2. Programming exercises

Do the following exercises in teams of 2 students. Use the English alphabet in the following exercises.

1. Design and implement a function to encipher a message m using the permutation cipher. Your function must receive as a parameter the plaintext m and a permutation π of size n . Your function must print the ciphertext c. **Use your functions of Section 1, to generate a random permutation**. Consider the case when the length of the message is not a multiple of n.

2. Design and implement a function to decipher a ciphertext c using the permutation cipher. Your function must receive as a parameter the ciphertext c and the permutation π of size n used to encipher . Use your function of Section 1 to compute the inverse permutation π^{-1}.

3. You must test your implementation in pairs as follows :

   a) One student will play the role of Alice and the other one will be Bob

   b) Then Alice must generate at random permutation π of size n. The size of the permutation must be given by the user. You must not need to modify the source code to establish it. Print the generated permutation.

   c) Alice must encipher a message in English using the permutation π previously generated with the permutation cipher. You must not need to modify the source code to establish the permutation or the message. The length of the message can be greater than n. Print the ciphertext c obtained.

   d) Alice will share with Bob, the permutation π and the ciphertext c obtained.

   e) Bob must decipher the ciphertext c. To do it, Bob will need to compute π^{-1}. Print π^{-1}. Then proceed to recover m and print it.

3. Products

Every student must write her/his own report, containing at least:

1. Personal information, date of the lab session and the topic that we are studying in this lab session.
2. List 10 permutations randomly generated and their corresponding inverse permutations.
3. Give 5 pairs of plaintext/ciphertext you used to encipher/decipher messages